Executive Summary

Retail Transaction Pre-Processing Pipeline

PURPOSE AND OUTCOME

The pipeline ingests simulated retail CSV transactions, standardizes field types, rejects invalid rows, and generates branch and customer-tier metrics. Of 14 input records, 10 passed validation and 8 valid, active transactions contributed to the sales results below.

14	71.4%	$1,356.49	36.2%
ROWS INGESTED	CLEAN RETENTION	GROSS SALES	MARGIN*

DATA INTEGRITY



10 accepted 4 rejected 2 valid but inactive (excluded from sales)

FINANCIAL SNAPSHOT

Profit before overhead*	**$491.49**
Estimated sales tax*	**$149.29**
Leading branch	**ON-101**
Leading branch sales	**$559.99**

CUSTOMER TIER BREAKDOWN

	COUNT	GROSS SALES
Platinum	2	**$650.00**
Gold	2	**$319.99**
Silver	2	**$279.50**
Bronze	2	**$107.00**

REJECTED RECORDS

- Negative revenue — CSV line 11
- Missing branch ID — CSV line 12
- Invalid calendar date — CSV line 13
- Unsupported region — CSV line 14

HOW THE PIPELINE WORKS

01 VALIDATE
Check required fields, dates, amounts, tiers and regions.

02 TRANSFORM
Filter active transactions; calculate tax and margins.

03 REPORT
Aggregate by branch and tier; print terminal summary.

INTERPRETATION AND LIMITATIONS

The sample indicates functioning validation and reporting, not real-store performance. Figures are based only on valid, active records; gross sales excludes estimated tax. Profit and margin exclude overhead. Regional tax rates are illustrative, and the 14-row synthetic dataset is too small to support operational forecasting.

SOURCE Included data/transactions.csv | 7 of 7 unit tests passed

* Calculated values are illustrative and are not audited financial or tax figures.